

POULTRY HOUSE HEALTH SYSTEM (PHHS)

The Poultry House Health System (PHHS) is an intelligent IoT-based monitoring and analytics platform designed to improve poultry health, environmental conditions, worker safety, and farm productivity through continuous monitoring of ammonia concentration and environmental parameters within poultry houses.

Powered by a low-energy nRF52 ARM Cortex-M4 processor and communicating via BLE, MQTT, and HTTP/HTTPS, the PHHS delivers real-time data to a mobile dashboard and cloud analytics engine — enabling proactive decision-making in commercial and precision poultry farming.

Key Features

- Continuous ammonia monitoring (1–500 ppm range)
- Temperature & humidity environmental sensing
- Poultry Health Index (PHI) analytics engine
- BLE communication for local wireless connectivity
- Cloud connectivity via MQTT and HTTP/HTTPS
- Mobile dashboard for real-time monitoring
- Configurable real-time alerts and notifications
- Low-power operation — USB or rechargeable battery

Technical Specifications

Ammonia Sensor	
Parameter	Value
Detection Range	1 – 500 ppm
Resolution	< 1 ppm
Accuracy	±5% of reading
Response Time	< 30 sec
Recovery Time	< 60 sec
Sensor Principle	Electrochemical

Temperature & Humidity	
Parameter	Value
Temperature Range	-10°C to +60°C
Temperature Accuracy	±0.5°C
Humidity Range	0 – 100% RH
Humidity Accuracy	±3% RH
Response Time	Instantaneous

Electrical Specifications	
Parameter	Value
Processing Unit	nRF52 ARM Cortex-M4
Supply Voltage	3.3V – 5V DC
Power Source	USB / Rechargeable Battery
Power Consumption (Active)	~80 mW
Power Consumption (Sleep)	< 5 mW
Battery Life (estimated)	> 72 hours continuous
Output Type	Digital (BLE, MQTT, HTTP/HTTPS)

Communication	
Parameter	Value
Wireless Protocol	Bluetooth Low Energy (BLE 5.0)
Cloud Protocol	MQTT / HTTP / HTTPS
Data Rate (BLE)	1 Mbps
Range (BLE)	Up to 30 m (line of sight)
Cloud Platform	Compatible with AWS, Azure, custom MQTT brokers

Environmental Ratings	
Parameter	Value
Ingress Protection	IP54
Operating Temperature	–10°C to +60°C
Storage Temperature	–20°C to +70°C
Operating Humidity	0 – 100% RH (non-condensing)
Dust Tolerance	Yes — tolerant of poultry house particulates

Pinout / Wiring	
Connector Pin	Function
VCC	3.3V – 5V DC power supply
GND	Ground
TX	UART Transmit (data output)
RX	UART Receive (configuration)

Mechanical Specifications	
Property	Value
Housing Material	ABS Polycarbonate
Protection Rating	IP54 (dust and splash resistant)
Dimensions	85mm (H) × 60mm (W) × 35mm (D)
Weight	~95 g
Colour	White / Light Grey
Mounting	Wall mount or pole clamp
Cable Connection	4-pin waterproof connector / USB-C

Applications

- Commercial poultry farms (broiler, layer, breeder houses)
- Precision poultry farming and data-driven flock management
- Worker safety monitoring — ammonia exposure alerts
- Smart agriculture and IoT-based environmental sensing networks
- Veterinary and research facilities

System Architecture

The PHHS follows a layered data pipeline from field sensing to cloud analytics:

Environmental Parameters	Sensors	Data Acquisition	Processing Unit	Analytics Engine	Alert Generation	BLE / Cloud	Mobile Dashboard
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